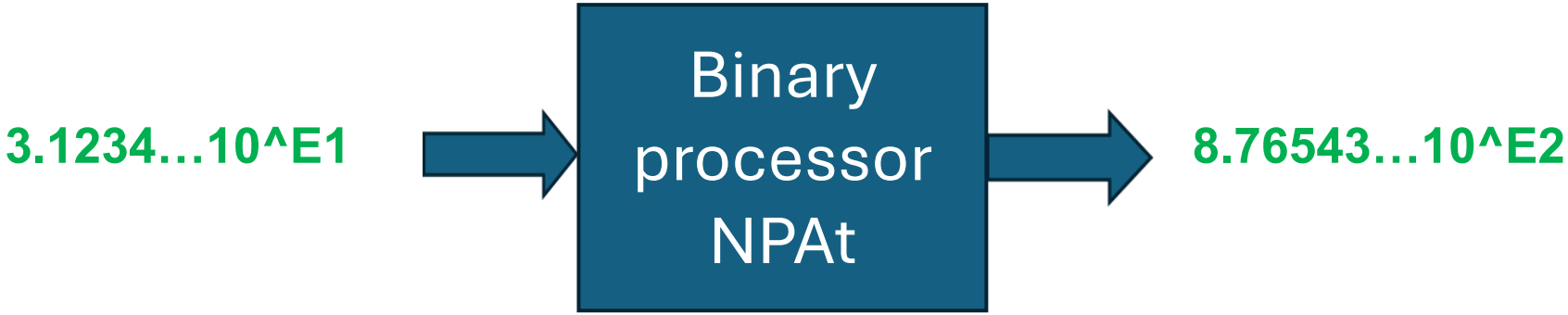
NPAt Core:
A New Paradigm in Binary Arithmetic

The Decimal-to-Binary Gap

- The current IEEE 754 standard is strictly binary-oriented.
- "Invisible" representation error occurs at the moment of decimal input.
- In long computational chains, this error leads to significant data distortion.

Redefining the Binary Processor



NPAt treats the binary processor as a precise instrument for decimal computations.

Goals:

- Maximum achievable number of reliable decimal digits in the output..
- Results that remain consistent with the original decimal data intent.
- High-speed processing without FPU overhead.

Accountability for Conversion Error

The Difference:

- IEEE 754: Ignores the initial representation discrepancy.
- NPAt: Explicitly accounts for the initial conversion error of non-representable decimal inputs.

Result:

- The calculation is mathematically anchored to the original decimal "truth" from step one.

Controlled Precision in Long Chains

- Maintains a controlled number of significant digits.
- Reliable for extensive sequences of arbitrary values.
- Prevents "money drift" in fintech and drift in scientific modeling.
- Tagline: Fintech-grade reliability at hardware speed.

Performance: 9x Faster

Why it works:

- **Optimal Utilization:** Maximizes standard CPU resources through the unique NPAt data structure.
- **Architectural Synergy:** Native optimization for 64-bit integer ALUs.
- **FPU-Free:** Bypasses subnormal numbers and mandatory normalization cycles.

Two Ways to Integrate NPAt

Pathway 1 – Compatibility Mode (IEEE 754)

This mode demonstrates the flexibility of the NPAt architecture by producing results identical to the current standard, but using superior logic.

Technical Verification:

Verified Integrity: Bit-for-bit identical results at I=1,000,000 addition steps.

Operational Synergy: This same mode operates faster on ALU, bypassing normalization cycles.

Primary Business Benefit:

Runs 9x faster than software emulation.

Microsoft Visual Studio Debu

```
i=      2 | NPAt=      914.913999999999987267074175179005 | IEEE=      914.913999999999987267074175179005
i=      3 | NPAt=      991.149999999999977262632455676794 | IEEE=      991.149999999999977262632455676794
i=      4 | NPAt=     1067.385999999999967258190736174583 | IEEE=     1067.385999999999967258190736174583
i=      5 | NPAt=     1143.622000000000070940586738288403 | IEEE=     1143.622000000000070940586738288403
i=      6 | NPAt=     1219.858000000000174622982740402222 | IEEE=     1219.858000000000174622982740402222
i=      7 | NPAt=     1296.094000000000278305378742516041 | IEEE=     1296.094000000000278305378742516041
i=      8 | NPAt=     1372.330000000000381987774744629860 | IEEE=     1372.330000000000381987774744629860
i=      9 | NPAt=     1448.566000000000485670170746743679 | IEEE=     1448.566000000000485670170746743679
i=     10 | NPAt=     1524.802000000000589352566748857498 | IEEE=     1524.802000000000589352566748857498
i= 100000 | NPAt=  7624362.441982096992433071136474609375 | IEEE=  7624362.441982096992433071136474609375
i= 200000 | NPAt= 15247962.441938884556293487548828125000 | IEEE= 15247962.441938884556293487548828125000
i= 300000 | NPAt= 22871562.442044571042060852050781250000 | IEEE= 22871562.442044571042060852050781250000
i= 400000 | NPAt= 30495162.442187622189521789550781250000 | IEEE= 30495162.442187622189521789550781250000
i= 500000 | NPAt= 38118762.442330673336982727050781250000 | IEEE= 38118762.442330673336982727050781250000
i= 600000 | NPAt= 45742362.442473724484443664550781250000 | IEEE= 45742362.442473724484443664550781250000
i= 700000 | NPAt= 53365962.442616775631904602050781250000 | IEEE= 53365962.442616775631904602050781250000
i= 800000 | NPAt= 60989562.442759826779365539550781250000 | IEEE= 60989562.442759826779365539550781250000
i= 900000 | NPAt= 68613162.442902877926826477050781250000 | IEEE= 68613162.442902877926826477050781250000
i=1000000 | NPAt= 76236762.443045929074287414550781250000 | IEEE= 76236762.443045929074287414550781250000
```

Pathway 2 – Native NPAt Core Strategy (Demo)

SThis mode, demonstrated by **NPAt_Demo**, is not just faster; it is fundamentally more stable.

Verification:

- **Test Case 1 (Summation):** Standard FPU logic accumulates visible error after only 10,000 steps of adding 0.1. NPAt remains mathematically accurate.
- **Test Case 2 (Subtractive Loss):** Standard double types lose critical significant digits due to catastrophic cancellation when processing values with high numeric contrast.
- **Data Integrity:** This demonstrates **controlled precision** in long calculation sequences of arbitrary values—a requirement that standard logic cannot meet without severe performance penaltie.

```
--- NPAt RESEARCH SUITE v3.2 [HARD-TEST] ---

====================================
N1 (Base): 0.1
N2 (Step): 0.1
z (Iter):  10000

--- COMPARISON (16 digits) ---
STD (Double): 1.000100000000159e+03
------------------------------------
NPAt (Core):  1.000100000000000E3 (q=1)

====================================
N1 (Base): 12345678.90987654
N2 (Step): -12345000.909
z (Iter):  10000

--- COMPARISON (16 digits) ---
STD (Double): -1.234376634110664e+11
------------------------------------
NPAt (Core):  -1.234376634110901E11 (q=1)

====================================
```

Verified Innovation & IP

Current Status:

U.S. Patent Pending (USPTO № 19/254,239).

Invention Title:

Apparatus for iteratively summing binary equivalents of decimal numbers.

Iouri Spiridonov

Credentials:

- **USSR State Prize Laureate.**
- Former Head of Laboratory at NPO "Agat", Belarus.
- 20+ years of R&D in development of digital systems.
- Recipient of the Medal for Labor Distinction.
- **Legacy:** 18 Invention Certificates and 2 prior in digital architecture.

Explore the NPAt Core

Action Items:

- Download the **NPAt_Demo** on GitHub.
- Review the TechRxiv preprints [1], [2].

Closing:

Open for professional discussion and licensing inquiries.